

# GOPOLANG MMUTLWANE

Data Science Graduate | Mathematics & Computer Science

Johannesburg, South Africa | pantsog24@gmail.com | 066 230 5349/073 617 2625 | gopolang-mmutilwane.onrender.com | github.com/gopolangmmutilwane

## PROFESSIONAL SUMMARY

BSc graduate in Mathematical Sciences (Mathematics and Computer Science) from the University of Johannesburg, currently completing a Postgraduate Diploma in Data Science (STADIO). Background combines formal training in statistics, linear algebra, and algorithm design with applied project work in classification, clustering, and relational database design. Focus has been on the full analytical pipeline rather than model output alone: identifying data-quality problems before modelling, evaluating models against baselines and business context, and documenting assumptions and limitations. Comfortable working across Python (scikit-learn, TensorFlow/Keras), SQL, and BI/reporting tools. Suited to graduate and junior roles in data science, data analytics, or data engineering.

## PERSONAL ATTRIBUTES

**Analytical & Critical Thinker • Detail-Oriented • Persistent Problem Solver • Self-Directed Learner • Adaptable • Quality-Conscious**

Evidence-driven in approaching analytical problems, with a strong emphasis on questioning assumptions, investigating data quality, recognising limitations, and communicating results responsibly.

## TECHNICAL SKILLS

- **Programming & Querying:** Python, SQL, C++, Java (academic coursework)
- **Data Analysis:** Pandas, NumPy, data cleaning, exploratory data analysis, feature engineering
- **Machine Learning:** scikit-learn, TensorFlow/Keras, logistic regression, SVM, decision trees, K-Means, hierarchical clustering, SHAP (model explainability)
- **Statistics:** Statistical inference, distribution theory, hypothesis testing, cross-validation
- **Visualisation / BI:** Matplotlib, Seaborn, Power BI, Streamlit
- **Databases / Data Engineering:** MySQL, dimensional (star schema) design, PostgreSQL/PostGIS
- **Cloud & Tools:** Git, GitHub, Jupyter Notebooks, VS Code, AWS S3 / EBS / EFS (AWS Certified Data Engineer Associate, in progress), BeautifulSoup, Requests

## EDUCATION

### Bachelor of Science in Mathematical Sciences: Mathematics and Computer Science

University of Johannesburg | 2025

- Relevant coursework: Statistical Inference, Distribution Theory, Linear Algebra, Real Analysis, Discrete Mathematics, Introduction to Numerical Analysis, Data Structures & Algorithms (C++), Object-Oriented Programming, Computer System Architectures

### Postgraduate Diploma in Data Science (NQF Level 8, In Progress)

STADIO | Commenced 2026

- Modules completed: Working with Data, Introduction to Data Science and Statistics, Data Science 1, Data Science 2

## SELECTED PROJECTS / TECHNICAL EXPERIENCE

**Loan Default Prediction: Neural Network Pipeline Debugging and Explainability** | Thuso Bank, STADIO coursework project (synthetic dataset)

**Objective:** Diagnose and correct an underperforming neural network pipeline predicting loan default on an imbalanced dataset (11% default rate).

**Contributions:** Debugged and rebuilt the TensorFlow/Keras pipeline; applied SHAP to explain which features drove individual predictions, illustrating how such models could support transparent credit decisions.

**Outcome:** Improved model performance to ROC-AUC 0.774 and average precision 0.302, versus a 0.107 baseline reflecting the class imbalance.

**Tools:** Python, TensorFlow/Keras, SHAP, Pandas

### **Customer Segmentation for Default Risk** | *Thuso Bank, STADIO coursework project (synthetic dataset)*

**Objective:** Determine whether customer segments could meaningfully predict default risk.

**Contributions:** Segmented customers using K-Means (K=4) and hierarchical clustering (Ward's linkage); tested segment membership against default outcomes.

**Outcome:** Found that segments did not meaningfully predict default risk and reported this as a negative result rather than force a positive conclusion.

**Tools:** Python, scikit-learn

### **Retail Sales Analytics Pipeline: Dimensional Data Warehouse** | *Personal project*

**Objective:** Build a dimensional data warehouse from a raw retail transaction dataset with no pre-existing schema.

**Contributions:** Designed a star-schema MySQL warehouse (fact\_sales plus supporting dimensions) from a 2,000-row, 3-category dataset; investigated data quality issues, including multi-valued customer attributes and invalid ages; documented modelling trade-offs such as keeping customer\_id as a degenerate dimension rather than building a separate customer table; produced accompanying requirements documentation, a data dictionary, and design-decision notes.

**Outcome:** Produced a working, documented relational schema with enforced referential integrity and traceable design decisions.

**Tools:** MySQL, SQL

### **Customer Churn Prediction: Mobile Telecom** | *Kopano Mobile, STADIO coursework project (IBM Telco Customer Churn dataset)*

**Objective:** Predict customer churn from telecom subscriber data.

**Contributions:** Investigated the dataset before modelling and identified 11 hidden missing values and a multicollinearity issue that would otherwise have compromised the model; built a logistic regression classifier.

**Outcome:** Achieved 80.3% accuracy, 0.646 precision, and 0.575 recall on the churn class.

**Tools:** Python, scikit-learn, Pandas

### **Purchase Intention Classification: E-Commerce** | *SmartShop Online, STADIO coursework project (UCI Online Shoppers Purchasing Intention dataset, 12,330 real sessions)*

**Objective:** Predict purchase intent from e-commerce session data.

**Contributions:** Compared logistic regression, SVM, and decision tree classifiers; used 5-fold cross-validation rather than a single train/test split to validate the comparison.

**Outcome:** Confirmed the decision tree's F1 advantage (mean 0.566) over the other models.

**Tools:** Python, scikit-learn

### **Booking Behaviour Prediction & Review Sentiment Analysis** | *British Airways, Forage job simulation*

**Objective:** Predict customer booking behaviour and assess passenger sentiment from review data.

**Contributions:** Built a predictive booking-behaviour model and communicated results through an automated python-pptx reporting pipeline; scraped and analysed passenger reviews for sentiment and anomalies, presenting findings through a Streamlit dashboard.

**Outcome:** Produced an automated reporting pipeline and an interactive sentiment-analysis dashboard.

**Tools:** Python, scikit-learn, python-pptx, Streamlit, BeautifulSoup, Requests

## **CERTIFICATIONS & PROFESSIONAL DEVELOPMENT**

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- AWS Certified Data Engineer Associate (DEA-C01), In Progress
- Data Science Job Simulation, British Airways (Forage), September 2025
- Introduction to Data Science Job Simulation, Commonwealth Bank (Forage), January 2026
- CHPC-NIThemCS Coding Summer School, Centre for High-Performance Computing & National Institute for Theoretical and Computational Sciences, February 2025